

TASK 1 — CTF CHALLENGE WALKTHROUGH

TryHackMe: Introductory Networking

1. Step-by-Step Account of What I Did

1. Opened the Introductory Networking room

I opened the TryHackMe Introductory Networking room and worked through the room from the Introduction section onward. The room covers the OSI model, TCP/IP model, practical networking concepts, and common networking tools.

2. Studied the OSI Model

I completed the OSI Model overview and reviewed the seven layers: Application, Presentation, Session, Transport, Network, Data Link, and Physical. This provided the layered framework used to understand how network communication works.

3. Worked through Encapsulation

I completed the Encapsulation section and followed how data is wrapped with the information required by lower network layers as it moves through the networking stack.

4. Studied the TCP/IP Model

I completed the TCP/IP Model section and compared the practical TCP/IP model with the more detailed OSI model.

5. Used Ping

I completed the Ping networking-tool section and used the tool to understand basic connectivity and response information.

6. Used Traceroute

I completed the Traceroute section and learned how traceroute can show the intermediate hops taken by a request on its way to a destination. I also completed the questions about selecting an interface and using TCP SYN requests.

7. Used WHOIS

I completed the WHOIS section and performed WHOIS lookups to understand domain-registration information such as registrant details, registration dates, nameservers, and related records.

8. Used Dig and studied DNS

I completed the Dig section and learned how DNS translates domain names into IP addresses. I also worked through questions covering recursive DNS servers, Top-Level Domain servers, the hosts file, public DNS servers, and TTL values.

9. Completed the room

I completed the Further Reading section and reached the final state showing the TryHackMe room as 100% completed.

2. What I Learned

OSI Model: The OSI model provides a seven-layer way of understanding network communication, with each layer handling a different part of the process.

TCP/IP Model: The TCP/IP model is a practical networking model used to describe how devices communicate across real networks.

Encapsulation: Network data is processed through layers, with each layer adding information needed for delivery and communication.

Ping: Ping is useful for checking whether a destination can respond and for observing basic round-trip timing.

Traceroute: Traceroute can reveal the sequence of intermediate network hops between a source and destination.

WHOIS: WHOIS can provide domain-registration information, although some personal information may be redacted.

DNS and Dig: DNS converts domain names into IP addresses. Dig can be used to query DNS records and inspect DNS responses.

TTL: A DNS TTL value represents how long a DNS result can be cached; 24 hours corresponds to 86,400 seconds.

3. Screenshot Evidence

10:16 PM TryHackMe | Introductory Networking
tryhackme.com

Room completed: 100%

Task 1 ✓ Introduction

The aim of this room is to provide a beginner's introduction to the basic principles of networking. Networking is a *massive* topic, so this really will just be a brief overview; however, it will hopefully give you some foundational knowledge of the topic, which you can build upon for yourself.

The topics that we're going to cover in this room are:

- The OSI Model
- The TCP/IP Model
- How these models look in practice
- An introduction to basic networking tools

Answer the questions below

Let's get started!

No answer needed ✓ Correct Answer

Task 2 ✓ The OSI Model: An Overview

Task 3 ✓ Encapsulation

Task 4 ✓ The TCP/IP Model

Task 5 ✓ Networking Tools Ping

Task 6 ✓ Networking Tools Traceroute

Task 7 ✓ Networking Tools WHOIS

Task 8 ✓ Networking Tools Dig

Task 9 ✓ Further Reading

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Task 1 Introduction

Task 2 The OSI Model: An Overview

The OSI (**O**pen **S**ystems **I**nterconnection) Model is a standardised model which we use to demonstrate the theory behind computer networking. In practice, it's actually the more compact TCP/IP model that real-world networking is based off; however the OSI model, in many ways, is easier to get an initial understanding from.

The OSI model consists of seven layers:

OSI:

Application

Presentation

Session

Transport

Network

Data Link

Physical

There are many mnemonics floating around to help you learn the layers of the OSI model -- search around until you find one that you like.

I personally favour: **A**nxious **P**ale **S**hakespeare **T**reated **N**ervous **D**runks **P**atiently

Let's briefly take a look at each of these in turn:

Layer 7 -- Application:

The  Application layer of the OSI model essentially provides networking options to programs running on a computer. It works almost exclusively with applications, providing an interface for them to use in order to transmit data. When data is given to the application layer, it is passed down into the presentation layer.



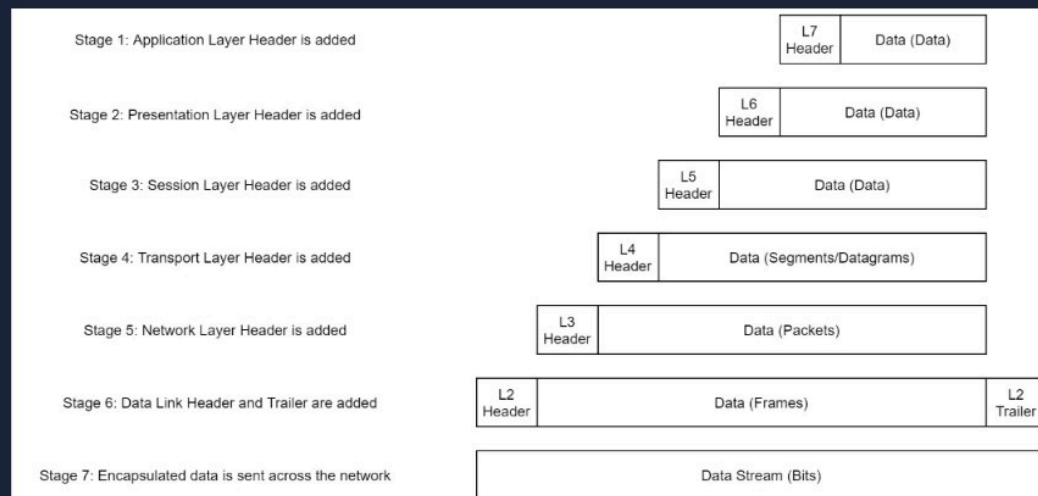


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Task 3 Encapsulation



As the data is passed down each layer of the model, more information containing details specific to the layer in question is added on to the start of the transmission. As an example, the header added by the Network Layer would include things like the source and destination IP addresses, and the header added by the Transport Layer would include (amongst other things) information specific to the protocol being used. The data link layer also adds a piece on at the *end* of the transmission, which is used to verify that the data has not been corrupted on transmission; this also has the added bonus of increased security, as the data can't be intercepted and tampered with without breaking the trailer. This whole process is referred to as *encapsulation*; the process by which data can be sent from one computer to another.



Notice that the encapsulated data is given a different name at different steps of the process. In layers 7, 6 and 5, the data is simply referred to as data. In the transport layer the encapsulated data is referred to as a segment or a datagram (depending on whether TCP or UDP has been selected as a transmission protocol). At the Network Layer, the data is referred to as a packet. When the packet gets passed down to the Data Link layer it becomes a frame, and by the time it's transmitted across a network the frame has been broken down into bits.

When the message is received by the second computer, it reverses the process -- starting at the physical layer and working up until it reaches the application layer, stripping off the added information as it goes. This is referred to as *de-encapsulation*. As such you can think of the layers of the OSI model as existing inside every computer with network capabilities. Whilst it's not actually as clear cut in practice, computers all follow the same process of encapsulation to send data and de-encapsulation upon receiving it.



Processes of encapsulation and de-encapsulation are very important -- not least because of their practical use, but also because they give us a standardised method for sending data. This means that all transmissions will consistently follow the same methodology, allowing any network enabled device to send





Room completed: 100%

Task 3 ✓ Encapsulation

Task 4 ✓ The TCP/IP Model

The TCP/IP model is, in many ways, very similar to the OSI model. It's a few years older, and serves as the basis for real-world networking. The TCP/IP model consists of four layers: Application, Transport, Internet and Network Interface. Between them, these cover the same range of functions as the seven layers of the OSI Model.

TCP/IP
Application
Transport
Internet
Network Interface

Note: Some recent sources split the TCP/IP model into five layers -- breaking the Network Interface layer into Data Link and Physical layers (as with the OSI model). This is accepted and well-known; however, it is not officially defined (unlike the original four layers which are defined in RFC1122). It's up to you which version you use -- both are generally considered valid.

You would be justified in asking why we bother with the OSI model if it's not actually used for anything in the real-world. The answer to that question is quite simply that the OSI model (due to being less condensed and more rigid than the TCP/IP model) tends to be easier for learning the initial theory of networking.

The two models match up something like this:

OSI	TCP/IP
Application	Application
Presentation	
Session	
Transport	Transport
Network	Internet
Data Link	Network Interface
Physical	



The processes of encapsulation and de-encapsulation work in exactly the same way with the TCP/IP model as they do with the OSI model. At each layer of the TCP/IP model a header is added during encapsulation, and removed during de-encapsulation.





Room completed: 100%

Task 5



Networking Tools

Ping



At this stage, hopefully all of the theory has made sense and you now understand the basic models behind computer networking. For the rest of the room we're going to be taking a look at some of the command line networking tools that we can use in practical applications. Many of these tools do work on other operating systems, but for the sake of simplicity, I'm going to assume that you're running Linux for the rest of this room. The first tool that we're going to look at will be the `ping` command.

The ping command is used when we want to test whether a connection to a remote resource is possible. Usually this will be a website on the internet, but it could also be for a computer on your home network if you want to check if it's configured correctly. Ping works using the ICMP protocol, which is one of the slightly less well-known TCP/IP protocols that were mentioned earlier. The ICMP protocol works on the Network layer of the OSI Model, and thus the Internet layer of the TCP/IP model. The basic syntax for ping is `ping <target>`. In this example we are using ping to test whether a network connection to Google is possible:

```
~$ ping google.com
PING google.com (216.58.198.174) 56(84) bytes of data.
```

Notice that the ping command actually returned the IP address for the Google server that it connected to, rather than the URL that was requested. This is a handy secondary application for ping, as it can be used to determine the IP address of the server hosting a website. One of the big advantages of ping is that it's pretty much ubiquitous to any network enabled device. All operating systems support it out of the box, and even most embedded devices can use ping!

Have a go at the following questions. Any questions about syntax can be answered using the man page for ping (`man ping` on Linux).

Answer the questions below

What command would you use to ping the bbc.co.uk website?

✓ Correct Answer

Ping *muirlandoracle.co.uk*

What is the IPv4 address?

✓ Correct Answer



What switch lets you change the interval of sent ping requests?

✓ Correct Answer





Room completed: 100%

```
9 ae28.londtt-sbr1.ja.net (146.97.33.61) 13.541 ms 13.229 ms 11.410 ms
10 72.14.205.74 (72.14.205.74) 15.143 ms 14.607 ms 13.865 ms
11 74.125.242.97 (74.125.242.97) 13.263 ms 74.125.242.65 (74.125.242.65) 12.553 ms 12.904 ms
12 172.253.71.191 (172.253.71.191) 13.943 ms 12.833 ms 172.253.71.189 (172.253.71.189) 12.631 ms
13 lhr48s23-in-f14.1e100.net (216.58.205.46) 13.227 ms 12.258 ms 12.482 ms
```

You can see that it took 13 hops to get from my router (`_gateway`) to the Google server at 216.58.205.46

Now it's your turn. As with before, all questions about switches can be answered with the man page for traceroute

(`man traceroute`).

Answer the questions below

Use traceroute on tryhackme.com

Can you see the path your request has taken?

✓ Correct Answer

What switch would you use to specify an interface when using Traceroute?

✓ Correct Answer



What switch would you use if you wanted to use TCP SYN requests when tracing the route?

✓ Correct Answer

[Lateral Thinking] Which layer of the *TCP/IP* model will traceroute run on by default (Windows)?

✓ Correct Answer

Task 7 ✓ Networking Tools WHOIS



Task 8 ✓ Networking Tools Dig



Task 9 ✓ Further Reading



How likely are you to recommend this room to others?

☐☐☐☐☐☐



Room completed: 100%

WHOIS lookup made at 02:22:04 07-Mar-2020

This is comparatively a very small amount of information as can often be found. Notice that we've got the domain name, the company that registered the domain, the last renewal, and when it's next due, and a bunch of information about nameservers (which we'll look at in the next task).

Your Turn

Answer the questions below

Perform a whois search on **facebook.com**

No answer needed

✓ Correct Answer

What is the registrant postal code for facebook.com?

94025

✓ Correct Answer

When was the facebook.com domain first registered (Format: DD/MM/YYYY)?

29/03/1997

✓ Correct Answer

Perform a whois search on **microsoft.com**

(Note: Please ensure you have read the task above before attempting the next questions.)

No answer needed

✓ Correct Answer

Which city is the registrant based in?

Redmond

✓ Correct Answer

[OSINT] What is the name of the golf course that is near the registrant address for microsoft.com?

Bellevue Golf Course

✓ Correct Answer

What is the registered Tech Email for microsoft.com?

msnhst@microsoft.com

✓ Correct Answer



Task 8 ✓

Networking Tools

Dig





Room completed: 100%

In the example will expire in two minutes and thirty-seven seconds.

Have a go at some questions about DNS and dig.

Answer the questions below

What is DNS short for?

Domain Name System

✓ Correct Answer

What is the first type of DNS server your computer would query when you search for a domain?

Recursive

✓ Correct Answer

What type of DNS server contains records specific to domain extensions (i.e. .com, .co.uk*, etc)*? Use the long version of the name.

Top-Level Domain

✓ Correct Answer

Where is the very first place your computer would look to find the IP address of a domain?

Hosts File

✓ Correct Answer

**[Research]** Google runs two public DNS servers. One of them can be queried with the IP 8.8.8.8, what is the IP address of the other one?

8.8.4.4

✓ Correct Answer

If a DNS query has a TTL of 24 hours, what number would the dig query show?

86400

✓ Correct Answer

Task 9 ✓ Further Reading



How likely are you to recommend this room to others?



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10:17 PM







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Task 6  Networking Tools Traceroute 

Task 7  Networking Tools WHOIS 

Task 8  Networking Tools Dig 

Task 9  Further Reading  

That's us completed our whirlwind tour of networking basics. Hopefully you've found it informative!

Networking is one of those things that you just need to learn. We've covered the very basics, but it would be a very good idea to continue to learn by yourself.

In terms of further information, feel free to reach out in the TryHackMe Discord if you want any help with the contents of this room. Additionally, if you want to expand your knowledge of networking theory, the [CISCO Self Study Guide by Steve McQuerry](#) is a great resource to work from. There may be a more up to date version available; however, this edition is cheap, readily available, and most importantly, still very relevant. Whilst it is designed to as a study guide for the CCNA exam, that book serves equally well as a very rounded introduction to networking principles.

Answer the questions below

Read the final thoughts

No answer needed

 Correct Answer

How likely are you to recommend this room to others?

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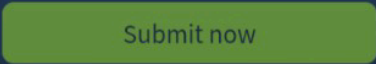
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4. Conclusion

I completed the TryHackMe Introductory Networking room and worked through the main networking concepts and tools covered in the tasks. The walkthrough demonstrates my understanding of the OSI and TCP/IP models, encapsulation, Ping, Traceroute, WHOIS, DNS, Dig, and TTL. The screenshots provide evidence of the work completed and the final completion of the room.